

VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELGAUM



Project phase -1(BIS685)

A Project Report on

“Development of A Secure Land Registration and Verification System”

A Dissertation submitted to the Department of Information Science and Engineering, RYMEC, BALLARI, affiliated by Visvesvaraya Technological University in the partial fulfillment of the Award of Degree of Bachelor of Engineering during the academic year 2025 = 2026.

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CERTIFICATE

This is to certify that the Project work entitled **“Development of A Secure Land Registration and Verification System”** is a Bonafide work carried out by Keerthi Maski (3VC23IS054), Honnuramma(3VC23IS037), NAME3(3VCXXISXXX) in partial fulfillment for the award of degree of Bachelor of Engineering in INFORMATION SCIENCE AND ENGINEERING from Visvesvaraya Technological University, Belgaum, during the year 2025-2026. The project report has been approved as it satisfies the academic requirements in respect of project work: Project phase - 1(BIS685) prescribed for Bachelor of Engineering degree.

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ABSTRACT

Land registration is a critical process for establishing legal ownership and preventing property disputes. However, traditional land registration systems often rely on paper-based records and centralized databases, which are vulnerable to document tampering, fraud, data loss, and unauthorized modifications. These issues can lead to ownership conflicts, delayed verification processes, and lack of transparency in property transactions.

The proposed **Development of a Secure Land Registration and Verification System** aim to provide a reliable and transparent digital platform for managing land ownership records. The system utilizes modern technologies such as **Blockchain** and geospatial verification to ensure secure storage, traceability, and authenticity of land-related information.

Blockchain technology is used to store land ownership details and transaction history in a distributed ledger, ensuring that records are immutable and resistant to tampering. Each land transaction is securely recorded and verified through cryptographic mechanisms, providing a transparent and trustworthy environment for property registration and transfer.

The system also incorporates **map-based verification** using geographic coordinates and digital mapping tools to accurately identify the location and boundaries of land parcels. This feature helps prevent duplicate registrations, boundary disputes, and fraudulent claims by linking land ownership data with verified geographic information.

A user-friendly interface enables authorized users and government officials to register land details, verify ownership records, view transaction history, and validate land locations through an integrated map visualization system.

This project aims to improve transparency, security, and efficiency in land registration processes while reducing the risk of fraud and disputes through the integration of blockchain technology and map-based land verification.

	CONTENTS		Page no
1	Introduction		1-1
2	Literature Survey		2-3
3	Objectives and Problem Statement		4-5
4	System Requirements		6-7
5	Basic Model/Architecture		7-9
6	Conclusion		10-14
7	References		15-15

1 Introduction

Land ownership is important for legal security and economic development. Traditional land registration systems mainly use paper-based records and centralized databases, which can lead to issues such as document tampering, fraud, lack of transparency, and delays in verification.

Blockchain technology offers a secure and reliable solution for managing land records. It provides a decentralized and tamper-resistant system where ownership details and transaction history can be stored safely. This improves transparency, data integrity, and trust among users.

The proposed system, Development of a Secure Land Registration and Verification System, uses blockchain technology along with map-based verification to manage land records digitally. The system helps in secure registration, accurate land identification, prevention of unauthorized modifications, and reduction of fraudulent transactions and boundary disputes.

By combining blockchain with digital map verification, the system aims to provide a transparent, efficient, and secure platform for land registration and ownership verification. Transactions more efficiently. Furthermore, the system promotes accountability and reduces the possibility of fraudulent property transfers.

2 Literature Review

Recent research has focused on using technologies such as Blockchain, Geographic Information Systems (GIS), and digital governance frameworks to improve the transparency, security, and efficiency of land administration systems. The recent literature can be summarized as follows:

1. QR Code and Blockchain for Land Verification (2024)

Singh et al. (2024) introduced a land registry system that combines blockchain with QR code verification to simplify land authentication and reduce fraud in property transactions. The use of QR codes allows quick access to land ownership details and improves verification efficiency.

2. Decentralized Land Registration Systems (2024)

Nair et al. (2024) developed a decentralized blockchain-based land registration model that improves transparency and prevents unauthorized modification of land records. Their system reduces dependency on centralized databases and strengthens trust in property transactions.

3. Blockchain for Secure Land Ownership (2023)

Shahariar et al. (2023) proposed a blockchain-based land record management system that ensures secure storage of land ownership data using cryptographic techniques. This approach also protects land records from manipulation and unauthorized access.

4. GIS-Based Land Parcel Mapping (2025)

Recent research highlights the use of GIS technologies for mapping land parcels and verifying boundaries through geo-tagging and spatial data analysis. GIS helps authorities accurately identify land locations and reduce boundary disputes.

5. Digital Land Administration Systems (2022)

Mezquita et al. (2022) reviewed digital land registry systems and concluded that blockchain-based solutions improve transparency, reduce corruption, and enhance efficiency in land management. Their study also emphasizes the importance of integrating digital technologies into traditional land administration frameworks

5. Blockchain-Based Land Registry Systems (2024)

and Twinomurinzi (2024) proposed a blockchain-based framework for land registration using Hyperledger Fabric. Their system ensures transparency, immutability, and secure management of land ownership records. This approach also enables trust information sharing between different government authorities involved in land Zein management

3 Objectives

The main objectives of the proposed system are:

- ❖ To develop a secure digital land registration system that ensures reliable storage and management of land ownership records.
- ❖ To implement **Blockchain technology** for maintaining an immutable and tamper-proof ledger of land transactions and ownership history.
- ❖ To design a transparent verification mechanism that allows authorized users and authorities to validate land ownership records easily.
- ❖ To integrate **map-based verification** using geographic coordinates and digital mapping tools to accurately identify land locations and boundaries.
- ❖ To reduce land-related fraud, duplicate registrations, and ownership disputes by ensuring secure and transparent record management.

To develop a user-friendly platform that enables landowners, buyers, and government authorities to register land, verify records, and track transaction history efficiently.

Problem Statement

Many land registration systems still rely on manual processes and centralized databases, which often lack security, transparency, and efficient verification mechanisms. These limitations increase the risk of document tampering, fraudulent ownership claims, and delays in property transactions.

In addition, the absence of proper map-based verification makes it difficult to accurately confirm land boundaries and ownership details. Therefore, there is a need for a secure and transparent digital system that can ensure reliable land registration and verification of property records.

4 System Requirements

Hardware Requirements

- ❖ Intel Core i5 Processor or equivalent
- ❖ 8 GB RAM
- ❖ 20 GB available Hard Disk Space
- ❖ Active Internet Connection

Software Requirements

1. **Smart Contracts:** Solidity
2. **Frontend:** HTML, CSS, JavaScript
3. **Backend:** Node.js / Express
4. **Map Verification:** Google Maps API
5. **Database:** MySQL or MongoDB
6. **IDE:** Visual Studio Code
7. **Blockchain Platform:** Ethereum / Hyperledger

2 Basic Model/Architecture

Implementation:

The system implementation consists of three main components:

- ❖ **Frontend:** Developed using HTML, CSS, and JavaScript to provide a user-friendly interface for land registration, map verification, and record viewing.
- ❖ **Backend:** Implemented using Node.js and Express to manage API requests, user authentication, and system operations.
- ❖ **Blockchain Module:** Smart contracts developed using Solidity are used to store land ownership records and transactions securely on the blockchain.

Advantages:

- ❖ Provides **secure and tamper-resistant** storage of land ownership records using blockchain technology.
- ❖ Enhances **transparency and traceability** in land transactions through decentralized record management.
- ❖ Reduces the possibility of **fraud, data manipulation, and unauthorized modifications**.
- ❖ Supports **accurate land validation** through map-based verification of geographical coordinates.
- ❖ Improves the **efficiency and reliability** of land record management systems.
- ❖ Enables **real-time verification** of land ownership and transaction history.
- ❖ Minimizes **human errors and administrative delays** in the land registration process.

- ❖ Provides a **transparent and trustworthy platform** for government authorities, buyers, and landowners.

Disadvantages

- ❖ Implementation of blockchain infrastructure requires higher initial setup cost and technical expertise.
- ❖ Integration with existing land record systems and government databases can be complex.
- ❖ Blockchain networks may introduce additional processing time for transaction verification.

The system depends on accurate geographic data for reliable map-based land verification

3 Conclusion

The proposed system **Blockchain-Based Land Registration System with Map Verification** offers a secure and transparent approach for managing land ownership records. By using blockchain technology, the system ensures tamper-resistant storage of land transactions through cryptographic security and decentralized data management, improving the reliability of ownership information.

The integration of map-based verification using geographic coordinates enables accurate validation of land locations and boundaries. This helps reduce duplicate registrations and minimizes landownership disputes.

Overall, the system enhances security, transparency, and efficiency in land record management while supporting reliable verification of property records for both government authorities and citizens.

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